Chethan S

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github.com/chethans2005 | chethan-s1122 | Portfolio | CLI

Profile

3rd-year Computer Science Engineering student at PES University with a strong focus on scalable AI systems and full stack development. Experienced in building end-to-end ML pipelines, AI-integrated web applications, and network-level systems; seeking Software Engineering roles where AI and scalability intersect.

Education

PES University, Bengaluru — 2023 – Present
B.Tech in Computer Science Engineering — CGPA: **8.66 / 10**

Gnana Dhama PU College — 2021 – 2023
PCMB (Science Stream) — Percentage: **94.5%**

Vidya Spoorti International Academy — 2016 – 2021
State Board — Percentage: **95.36%**

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, FastAPI, scikit-learn, XGBoost, LightGBM, SHAP, Optuna, python-pptx, Flutter

Databases: MongoDB, MySQL

Core Areas: Full Stack Development, Machine Learning, Deep Learning, AI Systems, Scalable Backends, Computer Networks

Tools & Technologies: Git, GitHub Actions (CI/CD), Linux, REST APIs, JWT, GetStream.io, Plotly, Uvicorn

Projects

Neko.ai – AI-Powered Presentation Generator `React | FastAPI | NLP` [Demo] [GitHub]

- Engineered a natural language to presentation pipeline using Llama-3.1-8B via Groq API, with Gemini 2.0 Flash as a fallback
- Architected a modular system for prompt parsing, slide generation, and professional/academic layout customization supporting up to 15 slides per request
- Enforced per-user rate limiting (50 requests max) and delivered real-time per-slide editing with python-pptx export

Neko-Matic – Configurable AutoML Platform `Next.js | FastAPI | ML` [GitHub]

- Developed a modular AutoML platform with a Next.js (TypeScript) dashboard and FastAPI backend supporting datasets up to 50 MB (∼10k rows)
- Automated model selection across 16 classifiers and regressors (Random Forest, XGBoost, LightGBM, SVM, and more) with Optuna-driven hyperparameter tuning
- Integrated SHAP for model explainability and Plotly/Seaborn for interactive leaderboard visualizations; surfaced top-5 model recommendations by default
- Designed async training pipelines with persistent run artifacts (model, pipeline, and training report) stored per run

LangBuddy – Real-Time Language Learning Platform `MERN Stack` [GitHub]

- Engineered a real-time chat platform with user matching across 14 language tags to connect learners by preference and proficiency
- Integrated GetStream.io for scalable messaging and implemented Google OAuth with JWT-based session management
- Crafted a fully responsive UI ensuring seamless interaction across desktop and mobile devices

Custom DHCP Client-Server System `Python | Networking | Security` [GitHub]

- Implemented a DHCP client-server system over UDP sockets, adhering to RFC 2131/2132 protocol specification
- Designed the full message lifecycle (DISCOVER, OFFER, REQUEST, ACK) with lease management via T1/T2 timers
- Secured packet transmission by integrating RSA-based digital signatures for authentication and tamper detection
- Deployed a multi-threaded server architecture to handle concurrent clients across Windows and Linux platforms

Leadership & Experience

Head of Operations – Equinox Club PES University | 2025 – Present

- Spearheaded execution of 3–5 technical events including a datathon and engineering challenges, reaching 100–200 participants
- Directed cross-functional teams across planning, logistics, and on-ground operations for each event

Certifications & Achievements

- **SDP-Cybersecurity Certification** – ExcelR Network security, threat analysis, and ethical hacking
- **Introduction to AppDev using Flutter & Dart** – PESU I/O
- **CNR Rao Scholarship** – PES University Awarded for academic merit